

Grade 8
Unit 8
Lesson 8 Practice

Name Answer Key
Date _____
Period _____

Consider the system of equations $\begin{cases} y = -2x + 5 \\ y = 0.25x - 4 \end{cases}$.

1. Complete the table to determine whether each ordered pair satisfies both equations.

Ordered Pair	$y = -2x + 5$	$y = 0.25x - 4$	Are both equations true?
$(-0.5, 6)$	$6 = -2(-0.5) + 5$ $6 = 1 + 5$ $6 = 6$ ✓	$6 = 0.25(-0.5) - 4$ $6 = -0.125 - 4$ $6 = -4.125$ ✗	☐ Yes ☒ No
$(-6, -5.5)$	$-5.5 = -2(-6) + 5$ $-5.5 = 12 + 5$ $-5.5 = 17$ ✗	$-5.5 = 0.25(-6) - 4$ $-5.5 = -1.5 - 4$ $-5.5 = -5.5$ ✓	☐ Yes ☒ No
$(4, -3)$	$-3 = -2(4) + 5$ $-3 = -8 + 5$ $-3 = -3$ ✓	$-3 = 0.25(4) - 4$ $-3 = 1 - 4$ $-3 = -3$ ✓	☒ Yes ☐ No

2. Based on your work for question 1, describe the solution set to the system of equations $\begin{cases} y = -2x + 5 \\ y = 0.25x - 4 \end{cases}$.

There is one solution:
 $(4, -3)$

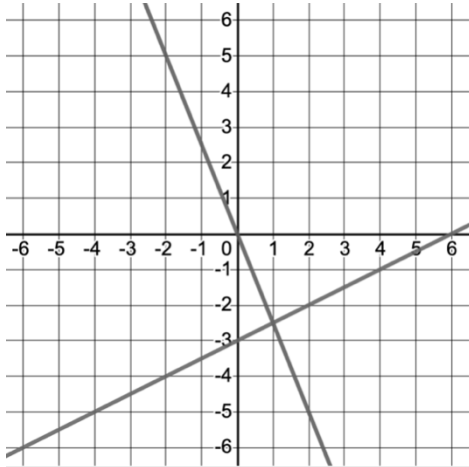
Consider the system of equations $\begin{cases} y = 0.25x - 2 \\ y = \frac{1}{4}x - 2 \end{cases}$.

3. Complete the table to determine whether each ordered pair satisfies both equations.

Ordered Pair	$y = 0.25x - 2$	$y = \frac{1}{4}x - 2$	Are both equations true?
$(-4, -3)$	$-3 = 0.25(-4) - 2$ $-3 = -1 - 2$ $-3 = -3$ ✓	$-3 = \frac{1}{4}(-4) - 2$ $-3 = -1 - 2$ $-3 = -3$ ✓	☒ Yes ☐ No
$(12, 1)$	$1 = 0.25(12) - 2$ $1 = 3 - 2$ $1 = 1$ ✓	$1 = \frac{1}{4}(12) - 2$ $1 = 3 - 2$ $1 = 1$ ✓	☒ Yes ☐ No

4. Based on your work for question 3, describe the solution set to the system of equations $\begin{cases} y = 0.25x - 2 \\ y = \frac{1}{4}x - 2 \end{cases}$. The system has an infinite number of solutions.

The system of equations $\begin{cases} y = -2.5x \\ y = 0.5x - 3 \end{cases}$ is graphed.



5. Does the system of equations have one solution, no solutions, or infinitely many solutions? If there is one solution, give the solution.

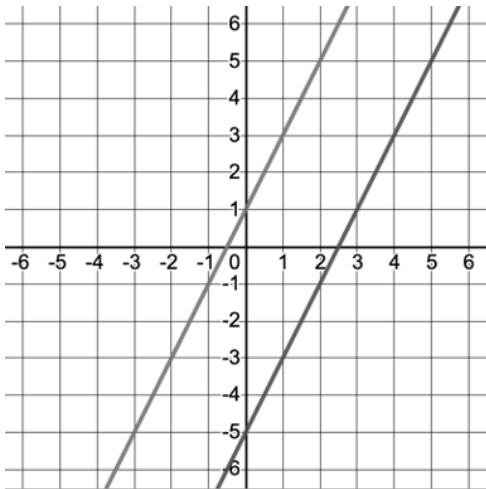
One solution
(1, -2.5)

6. Use substitution to show that the solution satisfies both equations in the system.

$$\begin{aligned} y &= -2.5x \\ -2.5 &= -2.5(1) \\ -2.5 &= -2.5 \quad \checkmark \end{aligned}$$

$$\begin{aligned} y &= 0.5x - 3 \\ -2.5 &= 0.5(1) - 3 \\ -2.5 &= 0.5 - 3 \\ -2.5 &= -2.5 \quad \checkmark \end{aligned}$$

The system of equations $\begin{cases} y = 2x + 1 \\ y = 2x - 5 \end{cases}$ is graphed.



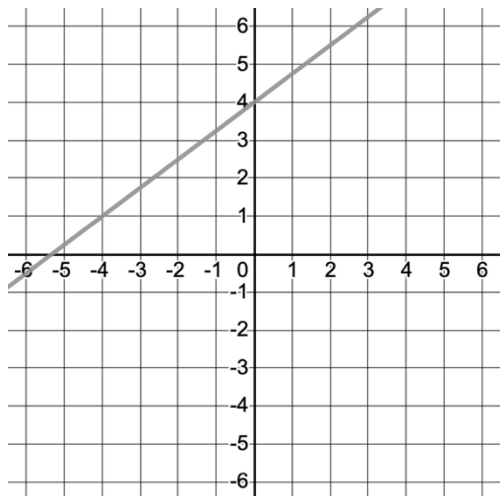
7. Does the system of equations have one solution, no solutions, or infinitely many solutions? If there is one solution, give the solution.

No solutions

8. Explain how you determined your answer to question 7.

The two lines have the same slope. This means they will never intersect, i.e. no solutions.

The system of equations $\begin{cases} y = 0.75x + 4 \\ y = \frac{3}{4}x + 4 \end{cases}$ is graphed.



9. Does the system of equations have one solution, no solutions, or infinitely many solutions? If there is one solution, give the solution.

Infinitely many solutions

10. Explain whether $(-4, 1)$ is a solution to the system of equations.

$(-4, 1)$ is on the line. So, it is one of many solutions to the system

$$\begin{aligned} 1 &= 0.75(-4) + 4 \\ 1 &= -3 + 4 \\ 1 &= 1 \quad \checkmark \end{aligned}$$